

# Cross-Modality Neonatal Brain Image Conversion Using a Latent Diffusion Model

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## I. ABSTRACT

Multi-contrast neonatal magnetic resonance imaging (MRI) is invaluable because T1-weighted (T1w) and T2-weighted (T2w) images provide different tissue properties. However, time and sedation constraints often preclude acquiring both imaging data. Here, we present a 2D slice-wise conditional latent diffusion model (LDM) that synthesizes T1w images from routinely collected T2w scans from neonates. Forty bias field-corrected T1w and T2w pairs were z-sliced and encoded by an Autoencoder-KL. Then, by concatenating spatially aligned conditioning information, conditional image-to-image translation generated synthetic T1w by inserting T2w latents, with a condition of MRI modality. On a held-out test set, the model achieved 21.03 dB peak signal-to-noise ratio (PSNR) / 0.6963 structural similarity index matrix (SSIM). We performed cross-modality conversion using LDMs, which enables to generate missing T1w with only T2w. Our approach provides a fast, sedation-free path to multi-contrast neonatal MRI and is readily extensible to infant or fetal data.

## II. INTRODUCTION

Magnetic resonance imaging (MRI) is an imaging technique that non-invasively depicts brain structure, and is especially useful for the developing brain. Multimodal MRI provides rich information to identify specific brain patterns in patients with neurological disorders as well as healthy individuals. T1-weighted (T1w) and T2-weighted (T2w) scans provide information regarding brain structure and are complementary. T1w emphasizes myelinated white matter and acute blood, whereas T2w highlights unmyelinated tissue and fluid accumulation [1]. Additionally, T1w/T2w intensity ratio is used to observe cortical microstructural profiles [2], in which clinicians can identify and diagnose the neurological disorders more efficiently. Collecting both contrasts in neonates, however, is problematic: scan times are tightly capped, motion artefacts are common, and pharmacological sedation carries non-trivial risks and cost [3]. Thus, missing either contrast undermines the diagnostic accuracy of multiple diseases.

One way to overcome this resource shortage is to synthesize the absent modality from the one that is routinely acquired. Early studies used generative adversarial network (GAN)-based cross-modality translation, but GANs often suffer from mode collapse resulted from local minimum, non-convergence and instability, because of the characteristics of training process [4, 5]. On the other hand, denoising-diffusion

probabilistic models (DDPMs) yield more stable and diverse outputs by learning a noise-perturbation process [6]. It appeared that DDPM is superior than GAN in image synthesis tasks [7], owing to its stochastic process in the training process. After continuously adding noises to input data, the pure noise image will be restored through reverse processes based on the Markov chain rule. It enabled diffusion-based model generate synthetic image approximate to target domain. DDPM has been applied to medical field and demonstrated its superior performance over GAN-based models [7]. However, these models require enormous memory through continuous denoising steps and computation power for 3-dimensional (3D) MRI data. Latent diffusion models (LDMs) alleviate this by compressing the input data to the latent space [8]. In brief, original images are compressed into low-dimensional space through the encoder, and it is restored to the original space through the decoder. The latent vectors following on the distribution obtained by the denoising stages are sampled. They are then decoded into original space and we can get synthetic image approximate to target domain without any priors. Here, by imposing conditioning key during diffusion process, we can transit the input image to another domain. Recent LDM variants, such as *Make-A-Volume* [9] and *CoLa-Diff* [10], already performed multi-modality conversion and demonstrated their effectiveness for synthesizing multimodal MRI in adults. However, the conversion in neonates remains largely unexplored. There is a difference between neonatal and adult brain, such as white matter (WM) and gray matter (GM) contrasts [11]. Therefore, image synthesis model for neonatal brains should be also considered to expand prior works on adulthood-focused multimodal synthesis models [13].

In this study, we introduce a 2D slice-based conditional LDM that synthesizes T1w from T2w scan in neonates. The proposed pipeline can provide multi-contrast information of neonatal brains in resource-constrained clinical settings.

## III. METHODS

### A. Participants and imaging data

We obtained imaging data of 40 neonates from the first release of the Developing Human Connectome Project (dHCP) database [14]. All participants were healthy term-born infants (37–44 weeks of age) with no major brain abnormalities. T1w and T2w data were acquired at the Evelina Neonatal Imaging Centre, London, using a 3T Philips Achieva scanner with a dedicated 32-channel neonatal head coil. T1w data were acquired using an inversion recovery fast spin-echo sequence (repetition time [TR] = 4,795 ms; inversion time [TI] = 1,740 ms; and echo time [TE] = 8.7 ms) with 0.8×0.8 mm in-plane resolution and 1.6 mm slice thickness. T2w images were

obtained using a fast spin-echo sequence (TR = 12,000 ms and TE = 156 ms) with the same resolution. Preprocessing steps included N4 bias field correction, skull-stripping, and tissue segmentation. 3D volumes (290×290×203 voxels) were sliced along the z-axis and grouped into triplets to form RGB-style tensors. This yielded 2,736 samples per modality. The dataset was randomly partitioned into training (1,936), validation (415), and test (416) sets (i.e., 70:15:15 split).

### B. Autoencoder

LDM was applied for generating synthetic images. It mainly comprises of three stages including compression, diffusion, and recovery processes. First, to learn compact and perceptually meaningful image representations, we trained an autoencoder that uses Kullback-Leibler regularization (AE-KL), which consisted of a convolutional layer-based encoder-decoder structure [15]. First, the encoder compressed T1w image (3×256×256) into a latent tensor (3×64×64), and the decoder reconstructed the data from the latent variable. Then, we computed the loss between actual input  $x$  and reconstructed image  $\hat{x}$ . This work was carried out in a way of reducing the reconstruction loss. Furthermore, AE-KL uses adversarial and KL losses [15]. For more efficient learning, the latent vector  $z$  was regularized to be zero-centered and obtain small variance using KL regularization parameter, which was then recovered into pixel-based space. Furthermore, AE-KL is trained in an adversarial manner like GAN [6]. The loss function ( $L_{AE-KL}$ ) used for training is as follows:

$$L_{AE-KL} = \min_{\mathcal{E}, \mathcal{D}} \max_{\psi} \left( L_{rec} \left( x, \mathcal{D}(\mathcal{E}(x)) \right) - L_{adv} \left( \mathcal{D}(\mathcal{E}(x)) \right) + \log D_{\psi}(x) + L_{reg}(x; \mathcal{E}, \mathcal{D}) \right), \quad (1)$$

where  $\mathcal{D}$  and  $\mathcal{E}$  denote decoder and encoder,  $L_{rec}$ ,  $L_{adv}$  and  $L_{reg}$  indicate reconstruction error, adversarial loss, and KL loss, and  $D_{\psi}$  is a patch-based discriminator. The autoencoder for T1w was trained for 200 epochs with  $\beta$  hyperparameter controlling the KL penalty, which was set to  $1e-6$ , and the best model was selected based on the lowest total validation reconstruction loss. We also performed AE-KL compression with T2w, which can be inserted as condition in the diffusion process. It was also performed for 200 epochs with  $\beta = 1 \times 10^{-6}$ , followed by the best model selected with lowest validation loss.

### C. Conditional latent diffusion model

To generate different modality images, we trained a conditional LDM. Specifically, it synthesizes T1w latents from T2w latents with a concatenation method. Using the frozen encoder/decoder from AE-KL, the LDM operates exclusively in the latent space. During the forward process, Gaussian noise was added iteratively to the T1w latent  $z_0$  over 1,000 steps to produce  $z_t$  in the latent space ( $z \in \mathbb{R}^3 \times 64 \times 64$ ). Diffusion process can be expressed as:

$$q(z_t|z_0) = \mathcal{N}(z_t | \alpha_t z_0, \sigma_t^2 \mathbb{I}), \quad (2)$$

starting from a latent vector  $z_0$  and defining a forward diffusion process  $q$ . Denoising diffusion formular are follows as:

$$p(z_0) = \int_z p(z_T) \prod_{t=1}^T p(z_t - 1|z_t), \quad (3)$$

where  $p(z_{t-1}|z_t)$  can be represented below:

$$\begin{aligned} p(z_{t-1}|z_t) &:= q(z_{t-1} | z_t, z_{\theta}(z_t, t)) \\ &= \mathcal{N}(z_{t-1} | \mu_{\theta}(z_t, t), \sigma_{t|t-1}^2 \frac{\sigma_{t-1}^2}{\sigma_t^2} \mathbb{I}) \quad (4) \end{aligned}$$

The reverse denoising process was learned via a time-conditional U-Net architecture. For paired image-to-image translation, we opted for concatenating methods conditioned by T2w latents,  $y$ . We compressed T2w image into latent tensor ( $\mathbb{R}^3 \times 64 \times 64$ ) same as T1w latents. For training, we used T1w-T2w pairs and concatenate both latent tensors, where T1w latents are injected as a condition [16]. Together, considering  $z$ -space, conditioning key and  $t$  time steps, the noise prediction loss ( $L_{LDM}$ ) was defined as follows:

$$L_{LDM} = \mathbb{E}_{z_0, y, \epsilon, t} [\|\epsilon - \epsilon_{\theta}(z_t, t, y)\|_2^2], \quad (5)$$

where  $z_t$  is the noisy latent at timestep  $t$ ,  $\epsilon \sim \mathcal{N}(0, 1)$  is sampled noise, and  $\epsilon_{\theta}$  is the denoising U-Net conditioned on  $y$ . Training lasted for 200 epochs with early stopping on the validation loss.

### D. Optimization strategy

This framework consisted of two stages: AE-KL and conditional LDM. After resizing T1w and T2w images to (265 × 265 × 3), each image was trained for 200 epochs. They shared the same optimizer parameters ( $\beta_1 = 0.5, \beta_2 = 0.9$  and a learning rate of  $4.5 \times 10^{-6}$ ). Then, the LDM was trained to generate T1w latent representations, with T2w latents used as conditions. T2w tensors were concatenated to provide conditioning key for U-Net. It was performed for 200 epochs with Adam optimizer ( $\beta_1 = 0.5, \beta_2 = 0.9$  and a learning rate of  $1 \times 10^{-6}$ ), and the batch size was 12. For T2w-to-T1w conversion, we sampled T1w tensors from the distribution using Denoising Diffusion Implicit Models (DDIMs), which had different conditions, transforming denoising steps with a strength parameter and guidance scale to compare the performance metrics and image quality in each case.

### E. Evaluation metrics

After choosing the final model based on the validation epoch loss, we evaluated the performance of the model with test data (416). The performance metrics are follows as: peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), mean absolute error (MAE), and normalized cross-correlation (NCC) between the actual and synthesized T1w.

## IV. RESULTS

### A. Synthesis performance

When we applied the proposed model to T2w scans of neonates, we could obtain the synthesized T1w images. To find out optimal conditions for t2-to-t1 modality transition, we performed sensitivity analysis following different combinations of hyperparameters, such as guidance scale, denoising strength with the number of DDIM steps fixed as 100. Not only quantitative aspects, but we also actually compared synthetic T1w images decoded and sampled from the distribution.

TABLE I. MODEL PERFORMANCE ON THE TEST SET

| Scale, Strength (%) | PSNR (dB) | SSIM    | MAE    | NCC    |
|---------------------|-----------|---------|--------|--------|
| 3.0, 50%            | 21.03,    | 0.6963, | 73.40, | 0.4713 |
| 3.0, 70%            | 21.02,    | 0.6966, | 72.95, | 0.4748 |
| 3.0, 80%            | 20.99,    | 0.6976, | 72.39, | 0.4725 |
| 3.0, 90%            | 20.91,    | 0.6985, | 71.28, | 0.4678 |
| 5.0, 50%            | 21.02,    | 0.6961, | 73.45, | 0.4715 |
| 5.0, 70%            | 21.03,    | 0.6967, | 72.92, | 0.4752 |
| 5.0, 80%            | 20.97,    | 0.6976, | 72.48, | 0.4714 |
| 5.0, 90%            | 20.92,    | 0.6988, | 70.93, | 0.4677 |
| 7.5, 50%            | 21.03,    | 0.6961, | 73.43, | 0.4718 |
| 7.5, 70%            | 21.02,    | 0.6966, | 72.96, | 0.4754 |
| 7.5, 80%            | 20.97,    | 0.6975, | 72.29, | 0.4728 |
| 7.5, 90%            | 20.93,    | 0.6985, | 70.94, | 0.4681 |

Abbreviations: PSNR, peak signal-to-noise ratio; SSIM, structural similarity index measure; MAE, mean absolute error; NCC, normalized cross-correlation. Scale, Guidance scale; Strength, Denoising Strength (%)

By visual inspection, the synthesized T1w from T2w showed comparable shapes of tissue boundaries (Fig. 1). Each of them indicates modality conversion results using the same LDMs, but with different hyperparameters for denoising stages. The further to the right, the higher the intensity, and the further down, the better the brain structure can be seen. We can identify brain structure more clearly following on how to set the hyperparameters in LDMs. Although several details of gray matter curvature need enhancements, the synthesized outcomes were overall comparable with the original images.

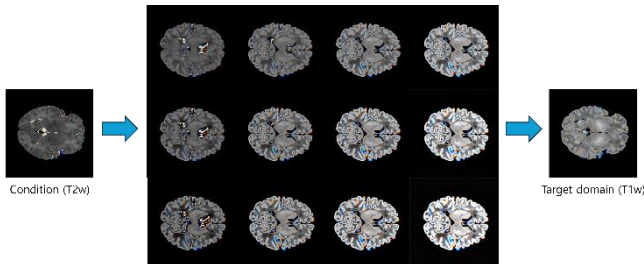


Fig. 1. Condition image (left), synthetic T1w following the settings (middle) and target domain (T1w). In the middle, there are several synthetic T1w according to the hyperparameters.

## V. DISCUSSION

In this study, we proposed a 2D slice-based LDM for generating T1w images from T2w for neonates. LDM enabled specific cross-modality transition by concatenating T2w as condition. Our work is one of the earliest applications of LDM to neonatal cross-modality synthesis for MRI, filling a gap left by adult-focused works. By synthesizing the missing T1w contrast from a single T2w scan, radiology teams can shorten acquisition times, avoid pharmacological sedation, and still access tissue contrasts essential for diagnosing multiple neurological disorders at early stages. However, several limitations should be acknowledged. A 2D-based approach may introduce z-axis discontinuities, and

thus, 3D perceptual studies are warranted. The number of participants was too small. In future studies, we will use the data from the full release of the database (~800 scans), and if available, we will collect multi-institutional datasets. Furthermore, we will test the generalizability of the model using the fetal MRI, which is also hard to obtain MRI for a similar reason as neonates. It can provide clinicians with sufficient information, followed by appropriate diagnosis for neurodevelopmental disease in pediatric population [18]. These directions aim to translate our model to a deployable tool in neonatal and broader pediatric neuroimaging.

## REFERENCES

- [1] Barkovich, A.J., Concepts of Myelin and Myelination in Neuroradiology. *American Journal of Neuroradiology*, 2000. 21(6): p. 1099-1109.
- [2] Soun, J.E., et al., Evaluation of neonatal brain myelination using the T1- and T2-weighted MRI ratio. *Journal of Magnetic Resonance Imaging*, 2017. 46(3): p. 690-696.
- [3] Dubois, J., et al., MRI of the Neonatal Brain: A Review of Methodological Challenges and Neuroscientific Advances. *J Magn Reson Imaging*, 2021. 53(5): p. 1318-1343.
- [4] Chen, H., Challenges and Corresponding Solutions of Generative Adversarial Networks (GANs): A Survey Study. *Journal of Physics: Conference Series*, 2021. 1827(1): p. 012066.
- [5] Isola, P., et al. Image-to-Image Translation with Conditional Adversarial Networks. 2016. arXiv:1611.07004 DOI: 10.48550/arXiv.1611.07004.
- [6] Ho, J., A. Jain, and P. Abbeel Denoising Diffusion Probabilistic Models. 2020. arXiv:2006.11239 DOI: 10.48550/arXiv.2006.11239.
- [7] Müller-Franzes, G., et al., A multimodal comparison of latent denoising diffusion probabilistic models and generative adversarial networks for medical image synthesis. *Scientific Reports*, 2023. 13(1): p. 12098.
- [8] Rombach, R., et al. High-Resolution Image Synthesis with Latent Diffusion Models. 2021. arXiv:2112.10752 DOI: 10.48550/arXiv.2112.10752.
- [9] Zhu, L., et al. Make-A-Volume: Leveraging Latent Diffusion Models for Cross-Modality 3D Brain MRI Synthesis. 2023. arXiv:2307.10094 DOI: 10.48550/arXiv.2307.10094.
- [10] Jiang, L., et al. CoLa-Diff: Conditional Latent Diffusion Model for Multi-Modal MRI Synthesis. 2023. arXiv:2303.14081 DOI: 10.48550/arXiv.2303.14081.
- [11] Xue, H., et al., Automatic segmentation and reconstruction of the cortex from neonatal MRI. *NeuroImage*, 2007. 38(3): p. 461-477.
- [12] Dubois, J., et al., The early development of brain white matter: A review of imaging studies in fetuses, newborns and infants. *Neuroscience*, 2014. 276: p. 48-71.
- [13] Kaplan, S., et al., Synthesizing pseudo-T2w images to recapture missing data in neonatal neuroimaging with applications in rs-fMRI. *NeuroImage*, 2022. 253: p. 119091.
- [14] Fitzgibbon, S.P., et al., The developing Human Connectome Project (dHCP) automated resting-state functional processing framework for newborn infants. *NeuroImage*, 2020. 223: p. 117303.
- [15] Kingma, D.P. and M. Welling Auto-Encoding Variational Bayes. 2013. arXiv:1312.6114 DOI: 10.48550/arXiv.1312.6114.
- [16] Zhan, Z., et al. Conditional Image Synthesis with Diffusion Models: A Survey. 2024. arXiv:2409.19365 DOI: 10.48550/arXiv.2409.19365.
- [17] Salimans, T., et al. Improved Techniques for Training GANs. 2016. arXiv:1606.03498 DOI: 10.48550/arXiv.1606.03498.
- [18] Banerjee, P., et al., Association Between Anesthesia Exposure and Neurocognitive and Neuroimaging Outcomes in Long-term Survivors of Childhood Acute Lymphoblastic Leukemia. *JAMA Oncology*, 2019. 5(10): p. 1456-1463.